

Name: _____

Class: _____

ACTIVITY SHEET

Chapter 3: Revision

1 Match each term with the appropriate description using an arrow.

a anion
b bonding pair
c cation
d chemical bond
e compound
f covalent bonding
g molecular compound
h network compound
i delocalised electrons
j directional bonding
k double covalent bond
l intermolecular bond
m intramolecular bond
n ion
o ionic bonding
p ionic compound
q metallic bonding
r multiple covalent bond
s non-bonding (lone) pair
t non-directional bonding
u polyatomic ion
v valence electrons
w valency

A attraction between cations and anions in an ionic compound
B attraction between delocalised valence electrons and metal cations
C attraction between protons and electrons of participating atoms
D attraction between shared electrons and nuclei of atoms involved
E bonding that occurs in most directions
F charged atom or group of atoms
G compound that is made up of positive ions and negative ions
H covalent bond with two pairs of electrons are shared
I covalent bond with more than one pair of shared electrons
J detached electrons that are free to move about within the metal
K direct line bonding between adjacent atoms in a molecule
L discrete molecule in which atoms are joined by covalent bonding
M electrons in the outer shell of an atom in its ground state
N electrostatic force of attraction that holds atoms together within a molecule
O electrostatic force of attraction between molecules in close proximity
P ions that consist of two or more atoms strongly covalently bonded
Q negatively charged atom or group of atoms
R pair of electrons shared by two atoms resulting in covalent bonding
S pair of valence electrons that are not involved in the covalent bond
T positively charged atom or group of atoms
U pure substance composed of different atoms bonded together
V number of single bonds with a H atom that the atom could form
W three-dimensional network of covalently bonded atoms

2 Explain the differences between the terms in each of the items below.

a Anions and cations

b Ionic, covalent and metallic bonding

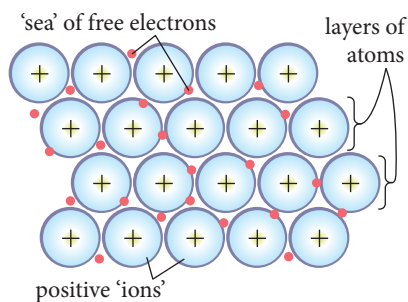
c Covalent molecular and covalent network compounds

d Directional and non-directional bonding

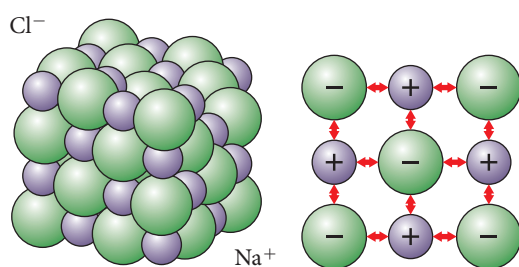
e Intermolecular and intramolecular bonds

3 Study each diagram and use it to describe the nature of that type of bonding.

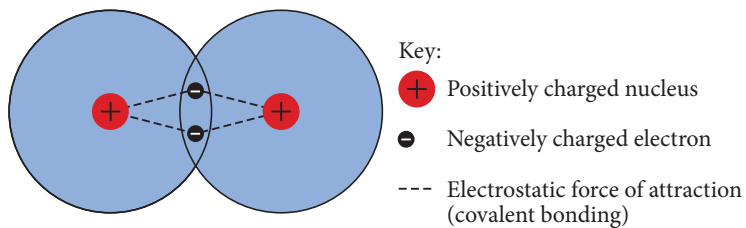
Metallic bonding



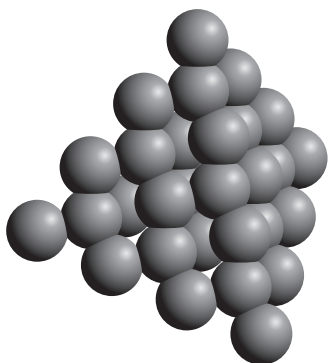
Ionic bonding



Covalent molecular bonding



Covalent network bonding



- 4 Use the model of **metallic bonding** that you described on the previous page to explain the following.
- a Metals are good conductors of electricity.
 - b Metals are malleable and ductile.
- 5 Use the model of **ionic bonding** that you described on the previous page to explain the following.
- a Ionic compounds have high melting points.
 - b Ionic compounds are hard but brittle.
- 6 Use the model of **molecular covalent bonding** that you described on the previous page to explain the following.
- a Molecular compounds have low melting and boiling points.
 - b Molecular compounds do not conduct electricity in the solid or liquid state.

7 Use the model of **network covalent bonding** that you described on page 3 to explain the following.

a Network compounds have very high melting points.

b The network compound graphite is a very good conductor of electricity.

8 Use the information in the table to determine the valency of the following.

Group number	1	2	3–12	13	14	15	16	17	18
Number of valence electrons	1	2	Variable	3	4	5	6	7	8
Valency	1	2	Variable	3	4	3	2	1	0

a Br ____

e Ar ____

i Si ____

b Ca ____

f H ____

j Al ____

c B ____

g S ____

k He ____

d P ____

h Na ____

l Cl ____

9 Complete this table summarising some properties of substances with different types of bonding.

Bonding \ Property				
	Do not conduct	Do not conduct (except for graphite)	Solids do not conduct but molten and aqueous states do	Good conductors in solid and liquid states
	Mostly soft	Hard and brittle	Hard and brittle	Hard and malleable
	Low	High	High	Variable but most commonly high
	Depends on solvent	Insoluble	Generally soluble in water	Generally insoluble

10 Use the information in the table that you completed above to identify the type of bonding likely to be found in each of the following substances.

a Baking soda _____

b Glass _____

c Steel _____

d Dry ice (CO₂) _____

- e Gold _____
- f Water _____
- g Table salt _____
- h Sugar _____

11 The table below summarises the physical properties of alkanes and alkenes.

Property	Alkanes and alkenes
State at 20°C	Lightest are gases
MPs and BPs	Both low
Conductivity	All non-conductors
Hardness	Solids are all soft
Malleability	Solids are malleable
Solubility	Insoluble in water

a What type of bonding is involved? Justify your answer.

c How strong would the intermolecular forces be?

d How strong would the intramolecular forces be?

12 Explain why alkenes can undergo addition reactions but alkanes cannot.